

A Universal Scaling Law for Ellipse Perimeters

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Abstract

Let

$$\mathbf{q}(\theta) = (a \cos \theta, b \sin \theta), \quad a \geq b > 0,$$

and let P_N be the perimeter of the inscribed polygon obtained from N equally spaced parameter values. We prove the exact factorization

$$P_N = \operatorname{sinc}\left(\frac{\pi}{N}\right) M_N, \quad \operatorname{sinc} x = \frac{\sin x}{x},$$

where M_N is the composite midpoint rule applied to the ellipse speed. Because that speed is periodic and analytic in a strip of half-width

$$\rho_* = \operatorname{artanh}(b/a),$$

the midpoint error is exponentially small for every fixed nondegenerate ellipse. Consequently,

$$P_N = P \operatorname{sinc}\left(\frac{\pi}{N}\right) + O(e^{-\rho N})$$

for every $0 < \rho < \rho_*$, and hence

$$P - P_N = P \left(\frac{\pi^2}{6N^2} - \frac{\pi^4}{120N^4} + \frac{\pi^6}{5040N^6} - \cdots \right) + O(e^{-\rho N}).$$

Thus the algebraic relative-error coefficients are universal: they do not depend on the ellipse axes. The tripling correction

$$\hat{P}_N = P_{3N} + \frac{P_{3N} - P_N}{8}$$

satisfies

$$P - \hat{P}_N = \frac{P\pi^4}{1080N^4} + O(N^{-6}) + O(e^{-\rho N}),$$

which gives fourth-order convergence after a single fixed-scale cancellation. The eccentricity dependence enters through the exponentially small quadrature remainder; as $b/a \rightarrow 0$, the analytic strip narrows like b/a , explaining why the asymptotic regime requires N of order a/b up to a tolerance-dependent logarithmic factor.

1 Introduction

Polygonal approximation of an ellipse contains two distinct numerical effects. First, replacing a smooth arc by its chord introduces a local chord–arc factor. Second, sampling the varying speed of the parametrization introduces a periodic quadrature error. Separating these effects yields a particularly simple scaling law.

The central result of this paper is ellipse-specific and exact: the equal-parameter polygonal perimeter factors into a universal sinc term and a composite midpoint approximation to the exact perimeter. Standard exponential convergence of periodic trapezoidal and midpoint rules then shows that, for every fixed $b > 0$, all algebraic powers of N^{-1} come from the sinc factor alone.

This statement is narrower and stronger than a blanket claim about arbitrary chord constructions on arbitrary smooth curves. It applies to the standard equal-parameter polygonal approximation of a nondegenerate ellipse.

2 Exact chord factorization

Let

$$h = \frac{2\pi}{N}, \quad \theta_j = jh, \quad t_j = \left(j + \frac{1}{2}\right)h, \quad j = 0, \dots, N-1.$$

Define the ellipse speed

$$v(t) = \|\mathbf{q}'(t)\| = \sqrt{a^2 \sin^2 t + b^2 \cos^2 t}.$$

The exact perimeter is

$$P = \int_0^{2\pi} v(t) dt = 4a E\left(1 - \frac{b^2}{a^2}\right),$$

where E is the complete elliptic integral of the second kind in the parameter convention.

Theorem 1 (Exact polygon–midpoint identity). *For every integer $N \geq 3$,*

$$P_N = 2 \sin\left(\frac{\pi}{N}\right) \sum_{j=0}^{N-1} v(t_j).$$

Equivalently, if

$$M_N = \frac{2\pi}{N} \sum_{j=0}^{N-1} v(t_j)$$

is the composite midpoint rule for P , then

$$P_N = \operatorname{sinc}\left(\frac{\pi}{N}\right) M_N.$$

Proof. For one chord, the angle-difference identities give

$$a \cos(\theta_j + h) - a \cos \theta_j = -2a \sin t_j \sin \frac{h}{2}$$

and

$$b \sin(\theta_j + h) - b \sin \theta_j = 2b \cos t_j \sin \frac{h}{2}.$$

Therefore its length is

$$\|\mathbf{q}(\theta_j + h) - \mathbf{q}(\theta_j)\| = 2 \sin \frac{h}{2} \sqrt{a^2 \sin^2 t_j + b^2 \cos^2 t_j} = 2 \sin \frac{h}{2} v(t_j).$$

Summing over j proves the first identity. Since $h = 2\pi/N$, the second follows by multiplying and dividing by h . $\square$

3 Analytic midpoint remainder

The factorization in Theorem 1 reduces the remaining problem to periodic quadrature.

Proposition 1 (Analytic strip). *Set $r = b/a$. For $0 < r \leq 1$, the function v has a 2π -periodic analytic continuation to every closed strip*

$$|\operatorname{Im} z| \leq \rho < \rho_*, \quad \rho_* = \operatorname{artanh}(r).$$

Proof. The possible branch points occur at zeros of

$$a^2 \sin^2 z + b^2 \cos^2 z = b^2 + (a^2 - b^2) \sin^2 z.$$

For $a > b$, the nearest zeros satisfy

$$\sin z = \pm i \frac{b}{\sqrt{a^2 - b^2}},$$

whose distance from the real axis is

$$\operatorname{arsinh}\left(\frac{b}{\sqrt{a^2 - b^2}}\right) = \operatorname{artanh}(b/a).$$

Hence a consistent analytic square-root branch exists in every smaller strip. When $a = b$, the speed is constant and the assertion is immediate. $\square$

Proposition 2 (Periodic midpoint error). *For each fixed ellipse with $a \geq b > 0$ and each $0 < \rho < \rho_*$,*

$$M_N - P = O(e^{-\rho N}).$$

Proof. The composite midpoint rule on a full period is a shifted periodic trapezoidal rule. For a periodic function analytic and bounded in a strip, its Fourier coefficients decay exponentially, and aliasing of those coefficients gives an exponentially small quadrature error. Applying this standard result to Proposition 1 proves the claim; see, for example, Trefethen and Weideman [2]. $\square$

4 Universal algebraic scaling law

Theorem 2 (Ellipse perimeter scaling law). *For any fixed ellipse $a \geq b > 0$ and any $0 < \rho < \rho_*$,*

$$P_N = P \operatorname{sinc}\left(\frac{\pi}{N}\right) + O(e^{-\rho N}).$$

Consequently, for every fixed integer $m \geq 1$,

$$P_N = P \sum_{\ell=0}^m \frac{(-1)^\ell \pi^{2\ell}}{(2\ell+1)! N^{2\ell}} + O(N^{-2m-2}) + O(e^{-\rho N}).$$

In particular,

$$P - P_N = P \left(\frac{\pi^2}{6N^2} - \frac{\pi^4}{120N^4} + \frac{\pi^6}{5040N^6} \right) + O(N^{-8}) + O(e^{-\rho N}).$$

Proof. Combine Theorem 1 with Proposition 2:

$$P_N = \text{sinc}\left(\frac{\pi}{N}\right) (P + O(e^{-\rho N})).$$

The sinc factor is bounded and has the Taylor series

$$\text{sinc } x = \sum_{\ell=0}^{\infty} \frac{(-1)^\ell x^{2\ell}}{(2\ell+1)!}.$$

Substitution gives the stated formulas. □

Remark 1. The word “universal” refers to the algebraic relative-error coefficients. For the standard equal-parameter ellipse polygon,

$$\frac{P - P_N}{P} \sim \frac{\pi^2}{6N^2},$$

independently of a and b . The onset of this asymptotic regime is not uniform as $b/a \rightarrow 0$.

5 Tripling-based cancellation

Define the one-step corrected perimeter

$$\hat{P}_N = P_{3N} + \frac{P_{3N} - P_N}{8} = \frac{9P_{3N} - P_N}{8}.$$

Corollary 1 (Fourth-order corrected law). *For every fixed nondegenerate ellipse and every $0 < \rho < \rho_*$,*

$$P - \hat{P}_N = \frac{P\pi^4}{1080N^4} - \frac{P\pi^6}{40824N^6} + O(N^{-8}) + O(e^{-\rho N}).$$

Hence

$$\frac{P - P_{3N}}{P - P_N} \rightarrow \frac{1}{9}, \quad \frac{P - \hat{P}_{3N}}{P - \hat{P}_N} \rightarrow \frac{1}{81}.$$

Proof. Apply the expansion in Theorem 2 at N and $3N$ and form $(9P_{3N} - P_N)/8$. The N^{-2} term cancels. The N^{-4} coefficient is $-P\pi^4/1080$ in $\hat{P}_N - P$, and the next coefficient is $P\pi^6/40824$. The ratio limits follow from the leading nonzero terms. □

This is a Richardson-type fixed-scale cancellation. It does not require knowledge of the perimeter or of the leading coefficient; only the known scale factor 3^{-2} is used.

6 Eccentricity and the asymptotic onset

The algebraic coefficients in Theorem 2 are independent of eccentricity, but the exponentially small midpoint remainder is not. Its natural strip width is

$$\rho_* = \text{artanh}(b/a).$$

For a highly eccentric ellipse,

$$\text{artanh}(b/a) \sim \frac{b}{a} \quad (b/a \rightarrow 0).$$

Thus suppressing the quadrature remainder to a fixed tolerance requires, at the level of exponential order,

$$N \gtrsim \frac{\log(1/\varepsilon)}{\operatorname{artanh}(b/a)} \sim \frac{a}{b} \log \frac{1}{\varepsilon}.$$

This explains an empirical inverse-axis-ratio scaling without asserting a universal finite- N threshold. The degenerate case $b = 0$ is excluded: there the speed becomes $a|\sin t|$, analyticity is lost, and the remainder is no longer exponentially small.

Ignoring the exponential transient, the leading relative-error design estimates are

$$N_{\text{raw}} \approx \frac{\pi}{\sqrt{6\varepsilon}}, \quad N_{\text{corr}} \approx \left(\frac{\pi^4}{1080\varepsilon} \right)^{1/4}.$$

They should be used only after $N \operatorname{artanh}(b/a)$ is sufficiently large.

7 Numerical verification

The accompanying script evaluates the exact perimeter using the complete elliptic integral and computes the polygonal sums directly at high precision. Table 1 displays the normalized quantities predicted by the theory.

b/a	N	$N\rho_*$	$N^2(P - P_N)/P$	$N^4(P - \hat{P}_N)/P$
0.90	162	238.46	1.6449031365	0.0901927055
0.50	486	266.98	1.6449306301	0.0901935031
0.10	1458	146.29	1.6449336850	0.0901935917
0.01	13122	131.22	1.6449340621	0.0901936027
Limit	—	—	$\pi^2/6 = 1.6449340668$	$\pi^4/1080 = 0.0901936028$

Table 1: Scaled relative errors for four axis ratios. The differing values of N place each case well inside its analytic asymptotic regime.

The data confirm two separate conclusions: the algebraic limiting constants are universal, while increasingly eccentric ellipses require larger N to make the exponentially small quadrature remainder negligible.

8 Scope and relationship to geometric N-series refinement

This theorem supplies an exact ellipse-specific foundation for the even-power assumption used in fixed-scale N-series acceleration. It is compatible with the general scaling-cancellation principle, but it does not assert that every chord-based construction on every smooth curve has the same coefficients.

The result is also distinct from nonlinear geometric residual iteration. Here the active variable is the resolution scale N^{-1} , and a linear combination of fixed-resolution approximants raises the algebraic order from two to four.

9 Limitations

The conclusions apply to:

- nondegenerate ellipses with $a \geq b > 0$;
- the standard polygon obtained from equal increments of the ellipse parameter;
- the asymptotic regime $N \rightarrow \infty$ for fixed axis ratio.

They do not claim:

- uniform exponential constants as $b/a \rightarrow 0$;
- the same sinc factorization for arbitrary parameterizations or meshes;
- numerical superiority over direct elliptic-integral algorithms;
- a universal theorem for all smooth closed curves without additional hypotheses.

10 Conclusion

The standard equal-parameter polygonal approximation of an ellipse satisfies an exact factorization:

$$P_N = \operatorname{sinc}\left(\frac{\pi}{N}\right) M_N.$$

For every fixed nondegenerate ellipse, periodic analytic quadrature makes $M_N - P$ exponentially small. The full algebraic error hierarchy therefore comes from the universal sinc factor, giving relative coefficients that are independent of the ellipse axes.

A single tripling-based cancellation removes the inverse-square term and produces fourth-order accuracy:

$$P - \hat{P}_N \sim \frac{P\pi^4}{1080N^4}.$$

Eccentricity controls not the algebraic coefficients but the onset of the asymptotic regime through the strip width $\operatorname{artanh}(b/a)$. This separation gives a rigorous and transparent scaling law for ellipse-perimeter N-series refinement.

References

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